



FROM WORKING PROTOTYPE TO A SAFE, SCALABLE LEARNING PLATFORM

Kitabu Digital production roadmap, requirements and launch checklist

Current position	Next decision gate
Interactive web and Android preview, role dashboards, offline-first concepts, paper viewer and curriculum review assets.	Approve a pilot school, launch subjects, legal owner, content rights and monthly budget before real student onboarding.

Prepared 1 October 2026 | Tanzania pilot edition

The decision in one page

Recommendation: move into a controlled production pilot, not a public launch. The interface is ready to demonstrate; identity, authorization, content licensing, operational security and real synchronization must be completed before real student data enters the system.

READY TO SHOW	NOT READY FOR REAL DATA	FOUNDER DECISIONS NEEDED
Landing page, student, teacher and parent previews; Android preview; in-app paper reading; calendar/timetable; role-aware floating tools.	Demo authentication, local-only state, incomplete tenant enforcement, no production monitoring or audited operating controls.	Pilot school, forms, launch subjects, legal entity/DPO, content permissions, cloud and AI budgets.

Production gate

- ☐ No real minors' data until PDPC registration, privacy notices and guardian handling are confirmed.
- ☐ No textbook or solution is published until rights and provenance are recorded.
- ☐ No SOC 2 claim until an independent auditor issues a report.
- ☐ No AI key in the browser or APK; all calls pass through the authenticated Kitabu server.

What has been completed

These items form a strong demonstration foundation. A checked item means implemented or verified in the prototype; it does not mean production certification.

- ☒ Public landing page and GitHub Pages demonstration
- ☒ Student, teacher and parent dashboard previews
- ☒ Parent-teacher conversation prototype
- ☒ Student progress, practice, examination and paper-reader flows
- ☒ School calendar and weekly timetable templates
- ☒ Retractable role-aware side dock; student tools hidden from parents
- ☒ Teacher-specific resource and assessment shortcuts
- ☒ Dark mode and responsive tablet/web layout
- ☒ Android preview package and in-app PDF reader
- ☒ Seven official TIE O-Level syllabi downloaded for private curriculum review
- ☒ NECTA/TETEA review catalogue with separate source provenance
- ☒ Paper ingestion and future OCR/camera service boundaries documented
- ☒ AI cost calculator and production-readiness plan
- ☒ Web build, unit tests and browser smoke test passing

What we need from you

Complete this checklist before production accounts and school onboarding are created. Keep credentials out of email, chat and GitHub.

Legal and privacy

- ☐ Registered legal name, BRELA/NGO certificate, TIN and business licence
- ☐ Audited financial statement if applicable
- ☐ Named Data Protection Officer and signed appointment letter
- ☐ Tanzanian legal review for child data and cross-border processing

Pilot school

- ☐ School name and authorized administrator
- ☐ Student, teacher and tablet counts
- ☐ Forms, subjects, start date and end date
- ☐ Tablet ownership and support responsibility
- ☐ Three measurable pilot success targets

Content

- ☐ Launch forms and subjects
- ☐ Named curriculum reviewer for each subject
- ☐ TIE/publisher permission to reproduce books
- ☐ Ownership permission for teacher-uploaded papers

Budget and ownership

- ☐ Company domain and email accounts
- ☐ Company-owned GitHub and cloud accounts
- ☐ Cloud monthly ceiling and AI monthly ceiling
- ☐ SMS/email/payment provider accounts

Content: useful without copyright risk

The library must distinguish permission from availability. "Free online" does not automatically grant the right to copy, modify, bundle offline or sell access.

Bucket	What Kitabu may do	Example
Licensed core	Store, download and use offline within the signed licence	TIE or publisher textbook under contract
Official reference	Link or use an internal review copy; do not redistribute	TIE curriculum or syllabus
Open supplementary	Download only when the exact edition has a compatible licence and attribution	Verified CC-licensed maths/science material
Unverified	Quarantine until ownership and permission are confirmed	Random PDF or unreviewed teacher upload

Safe publishing pipeline

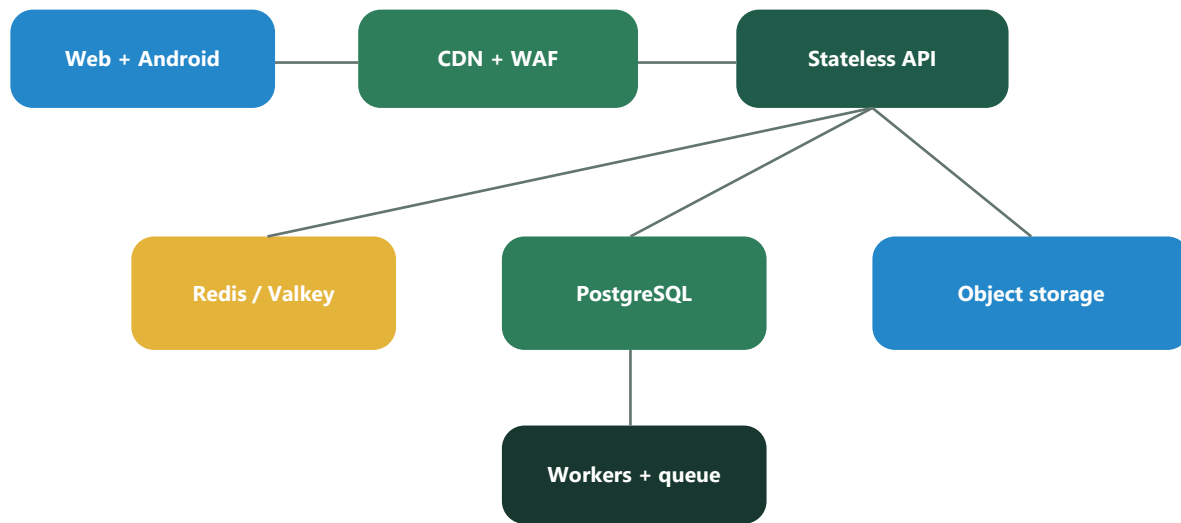
1 Acquire Source + owner	2 Verify Licence + hash	3 Map TIE competency	4 Extract OCR + layout	5 Review Two educators	6 Publish Version + withdraw
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Already collected for private review: Mathematics, Physics, Chemistry, Biology, English, Kiswahili and Geography O-Level syllabi. Each file has a source URL, owner, file hash and a redistribution-blocked status.

Important: third-party worked solutions must never be labelled as official NECTA marking schemes.

Scale without overbuilding

Start with managed services and measured scaling. The database remains the source of truth; caching reduces repeated work but does not replace authorization or durable records.



Cache rules

- ☐ Use cache-aside reads with short TTLs and versioned invalidation.
- ☐ Prefix every key with school/tenant and permission context.
- ☐ Cache public curriculum metadata and computed dashboard summaries.
- ☐ Do not cache passwords, OTPs, unrestricted chat history or full student records.
- ☐ Use Redis/Valkey for rate limits, sessions, temporary sync state and job queues.

Indicative pilot infrastructure

Component	Starting approach	Indicative monthly allowance
Web/API	Managed container platform	\$25-\$75
Database	Managed PostgreSQL with backups	\$20-\$80
Cache/jobs	Managed Valkey/Redis	\$15-\$30
Files/CDN	Private object storage	\$5-\$25
Monitoring/email/SMS	Usage dependent	\$15-\$40+
Working pilot range	Before AI, staff and legal costs	\$80-\$250+

Security, privacy and SOC 2

Current status: prototype only. Kitabu is not SOC 2 compliant and must not describe itself as certified. SOC 2 is an independent auditor report against the AICPA Trust Services Criteria, not a cloud setting.

Control area	Production requirement	Status
Identity	Verified login, MFA for staff, rotated refresh tokens, revocation	Build next
Authorization	School, class, guardian-child and teacher-class enforcement in API	Build next
Data protection	TLS, encrypted storage/backups, secrets vault, key rotation	Build next
Uploads	Type validation, malware scan, quarantine, human approval	Build next
Operations	Monitoring, alerts, audit logs, restore drills, incident response	Build next
Privacy	PDPC registration, DPO, notices, retention, deletion/export	Founder + legal
Assurance	Penetration test, control evidence, readiness review, audit	Before claims

SOC 2 path



Start with the Security criteria. Operate the controls long enough to produce evidence, complete a readiness review, then engage a licensed audit firm for Type I and later Type II. Add Availability, Confidentiality or Privacy according to contractual commitments.

Tanzania: appoint a DPO and complete PDPC registration before collecting real student information. Confirm the applicable controller/processor roles with Tanzanian counsel.

AI tutor: controlled, cited and affordable

The tutor should retrieve from an approved, versioned curriculum corpus instead of treating the open internet as the syllabus. It should cite the lesson or source, expose uncertainty, and use authored rubrics for mathematics and science.

ROUTINE TURN	DIFFICULT TURN	EXAM MODE
Small, economical model for hints, explanations and lesson retrieval.	Escalate only complex reasoning or low-confidence requests to a stronger model.	Disable hints and answers. Preserve test integrity and record submission events.

Illustrative text-only estimate

Assumption	Value
Students	500
Tutor turns per student/month	40
Average input/output	1,800 / 450 tokens
Prompt cache hit assumption	60%
Estimated generation cost	\$12.31/month
With 30% contingency	\$16.01/month
Recommended closed-pilot ceiling	\$50/month with alerts and kill switch

This excludes OCR, embeddings/vector storage, retries, moderation where billable, cloud hosting, taxes and support. Measure at least 1,000 representative conversations before setting the production budget. Never ship the AI provider key in web code or the APK.

Twelve-week controlled production path



Weeks 1-2 - Decisions and controls

Confirm legal entity, DPO, pilot school, subjects, rights strategy, cloud ownership, threat model and budgets.

Weeks 3-6 - Production foundation

Build real identity/RBAC, tenant isolation, API integration, storage/upload scanning, backups, logs and alerts.

Weeks 7-10 - Learning operations

Complete offline sync, reviewed curriculum/questions, teacher tools, AI evaluation harness and staff-only trial.

Weeks 11-12 - Release gate

Run load test, penetration test, restore drill, privacy review, Android signing and a documented go/no-go review.

Launch acceptance checklist

- ☐ PDPC and privacy requirements signed off
- ☐ Every production role passes authorization tests
- ☐ Backup restoration succeeds within the agreed recovery time
- ☐ Licensed content register is complete
- ☐ AI spending limits and safety evaluations pass
- ☐ Pilot school staff complete training
- ☐ Support, incident and content-withdrawal owners are on call

Sources and working files

Primary references used for this roadmap. Prices and regulations can change; verify them again before signing contracts.

Tanzania Institute of Education - lower secondary syllabi

<https://tie.go.tz/publications/syllabus-for-lower-secondary-academics>

Personal Data Protection Commission - registration

<https://pdpc.go.tz/services/registration/>

PDPC - registration requirements and fee guidance

https://pdpc.go.tz/feedback-faqs/?faq_page=2

AICPA - SOC suite and Trust Services Criteria

<https://www.aicpa-cima.com/resources/landing/system-and-organization-controls-soc-suite-of-services>

OpenAI API pricing

<https://platform.openai.com/pricing>

DigitalOcean App Platform pricing

<https://docs.digitalocean.com/products/app-platform/details/pricing/>

DigitalOcean managed database pricing

<https://www.digitalocean.com/pricing/managed-databases>

DigitalOcean Spaces pricing

<https://docs.digitalocean.com/products/spaces/details/pricing/>

Project references

Production checklist: docs/PRODUCTION-READINESS.md

Paper ingestion contract: docs/architecture/paper-ingestion-contract.md

Curriculum review manifest: artifacts/tanzania-content-review/manifest.json

AI cost calculator: scripts/estimate-ai-cost.mjs

Final note: do not set a public production date until content rights, PDPC registration, role authorization and restore testing are complete.